Alizainul Vaghjipurwala, M.Sc.

Mechatronics Engineer

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Summary

Having worked extensively on **Sensor Fusion** and **Machine Learning**, I love solving complex challenges by identifying how to collect the right data from the available resources, extract the most information/knowledge out of it, and get optimum use by making models for fulfilling a wide variety of tasks, from prediction to generation.

Work Experiences

Volkswagen AG

PhD Researcher — Braunschweig, Germany — 09/2025 - Present

- Developing a **Data Driven - Condition Monitoring and Predictive Maintenance framework** for safety critical Steer by Wire system, integrating **Signal Processing, Sensor Fusion** and **Machine Learning.**
- Fault analysis through **FMEA** datasheets.
- Development of system simulation environment in **Simulink**.
- Identifying the signals based on Monotonicity, Prognosability, Trendability as well as ability to isolate fault effects.
- Using Sensor / Data Fusion technique for tackling challenges of fault/signal observabilty in safety critical mechatronical systems.
- Investigating **explainable ML Techniques** (ex. SHAP Analysis) to identify signal and component contribution.

Hella Forvia GmbH

Research Engineer (Master Thesis) — Lippstadt, Germany — 02/2024 - 08/2024

- **Thesis Title**: Ego-vehicle self-localization and trajectory identification using **GPS-IMU sensor-fusion, Stereo Cameras** and **OpenDRIVE maps**.
- Designed a **Deep Learning** -based **virtual sensor** framework to simulate GPS data during signal loss, enhancing localization robustness in challenging environments.
- Built trajectory prediction model, using vehicle telemetry and road topology data, enabling more intelligent path estimation under uncertainty.
- Developed method to build lane level constraints from map data to enhance localization robustness.
- Applied **signal processing** and ML techniques for sensor data analysis to detect patterns and anomalies in driving behavior.

Hella Forvia GmbH

Engineering Intern — Lippstadt, Germany — 08/2023 - 02/2024

- Contributed to the development of **3D object detection** and **tracking** using **YOLOv8**, as well as Lane detection for improved perception accuracy.
- Developed automated map parsing system for converting OpenDRIVE map data into MATLAB structures, enabling rapid prototyping.
- Built road network graphs using data from OpenDRIVE maps.
- Calibrated stereo and monocular cameras using Chessboard images.

Sodecia India Pvt. Ltd.

Graduate Engineer Trainee — Halol, Gujarat, India — 07/2019 - 10/2019

- Planning and optimization of production by analyzing inventory, requirements and sales data.

Skills

Python | Matlab / Simulink | C++ | Pytorch | TensorFlow | OpenCV | Scikit-learn | Numpy | Pandas | Matplotlib | PCL | Keras | ROS | GIT | AWS | MLOps | Docker | Github | GNSS / GPS | IMU | Cameras (Mono, Stereo) | LiDAR

Analytical Thinking | Communication | Presentation | Team work | Curiosity | Quick learner | Dashboards

Education

M.Sc. Mechanical Engineering ISE (Mechatronics Profile) — GPA - 1.9

Universität Duisburg Essen — Duisburg, Germany — 10/2021 - 08/2024

- Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning
- Control Theory, Model Predictive Control, Vision-Based Control
- Mathematical Optimization, Signal Processing, Kalman & Particle Filters
- Kinematics & Dynamics of Robotic Systems, Sensor Fusion

B.E. Mechanical Engineering

Gujarat Technological University — Gujarat, India — 09/2015 - 06/2019

- Bachelor thesis: City Efficient Cars - Redesigning cars for reducing space occupied in the cities while in idle/parked condition

Highschool (Qualification for University Education)

Calorx Public School — Godhra, India — 07/2013 - 03/2015

Core Expertise

Multimodal Sensor Fusion, Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Large Language Models (LLMs), Natural Language Processing (NLP), ML Pipeline Design, TensorFlow, PyTorch, Containerization, Transformers.

Languages

- **German** - Upper Intermediate (B2)
- **English** - Native/Bilingual (C1)
- **Hindi** - Native

Certifications and Courses

The Complete Self Driving Car Course - Applied Deep Learning

UDEMY

- Deep Neural Networks, Convolution Neural Networks, Multiclass Classification, Image recognition, Polynomial Regression, leNet, Data Augmentation, Object Detection.

Deep Learning with MATLAB

Mathworks MATLAB

- Deep Network Design, Image Classification, Deep Learning for Computer Vision, Regression, Sequence Classification, Data Processing, Pattern Identification.

Reinforcement Learning with MATLAB

Mathworks MATLAB

- Deep Reinforcement Learning, Q-Learning, Deep Q-Networks (DQN), Agent-Environment Interaction, State, Action, Reward, Action-Value Function (Q).

Volunteering

Social worker, Head coordinator

National Service Scheme (NSS), India

- Lead a team of 70 volunteers, organizing various events for underprivileged children, focussing on their education and overall development.
- Educating people in rural regions, farmers and dairy workers about the various government schemes.

Volunteer

Share A Book India Association (SABIA), India

- Book donation drives to build libraries in Government Schools in rural areas.

Volunteer

Campus Garten, Germany

- Community gardening at the university campus along with sustainabiliy programs like setting up rain water harvesting, organising clean-up drives and other events and discussions.

Hobbies and Interests

AI | Physics | Basketball | Quizzing | Volunteering

Personal Projects

GitHub - AliVaghjipur

LiDAR-Camera Fusion for 3D Object Detection

- Implemented state-of-the-art sensor fusion pipeline using **KITTI dataset**, achieving 89% detection accuracy.
- Integrated **YOLOv5** with point cloud processing for robust **3D object detection** and **tracking**.
- Developed custom data preprocessing pipeline handling multi-sensor synchronization.

Advanced Lane Detection

- Applied Sobel and color thresholding with CLAHE for robust feature extraction under varying lighting.
- Implemented sliding window and memory-based lane tracking for frame-to-frame stability.
- Estimated lane curvature and vehicle offset in real-world units; overlayed results on video frames.
- Enhanced robustness with ROI masking and temporal smoothing for challenging environments (e.g., shadows, glare, tight curves).

Stable Diffusion model from scratch

- Built a Stable Diffusion model from scratch using **PyTorch**, building a complete **text-to-image** and **image-to-image Transformer based** generation pipeline, gaining deep insights into **generative AI** models.
- Developed all components like the **Variational Autoencoder (VAE)** for latent encoding, a transformer-based text encoder and a U-Net with attention for denoising, CLIP and DDPM scheduler, developing complete end-to-end generative AI model.

Boston Housing Price prediction (model + deployment)

- Developed and deployed a Boston Housing Price Prediction web app using Flask, HTML, and Postman for API testing, serving a pickled Linear Regression model.
- Containerized the application with **Docker** and automated deployment using **GitHub Actions** and Render/Heroku, ensuring seamless **CI/CD**.
- Built and trained the model on the Boston Housing dataset, implementing data preprocessing, feature engineering, and model serialization for production readiness.

Autonomous Vehicle Behavioral Cloning

- Developed end-to-end deep learning system for autonomous vehicle control using behavioral cloning.
- Implemented **data augmentation** pipeline improving model generalization by around 35%.
- Achieved 97% successful autonomous navigation in various weather and lighting conditions.